

Lab 3 Solutions

Consider the following optimization problem:

$$\text{Minimize } J(\theta_1, \theta_2) = 5\theta_1^2 + 3\theta_2^2$$

subject to the constraint:

$$4\theta_1 + 2\theta_2 = 12$$

The Lagrangian function combines the objective function and the constraint using a Lagrange multiplier λ . The Lagrangian is defined as:

$$L(\theta_1, \theta_2, \lambda) = 5\theta_1^2 + 3\theta_2^2 + \lambda(4\theta_1 + 2\theta_2 - 12)$$

To find the stationary points, we compute the gradients of the Lagrangian with respect to θ_1 , θ_2 , and λ , and set them equal to zero:

$$\frac{\partial L}{\partial \theta_1} = 10\theta_1 + 4\lambda = 0 \quad \Rightarrow \quad \theta_1 = \frac{-2\lambda}{5}$$

$$\frac{\partial L}{\partial \theta_2} = 6\theta_2 + 2\lambda = 0 \quad \Rightarrow \quad \theta_2 = \frac{-\lambda}{3}$$

$$\frac{\partial L}{\partial \lambda} = 4\theta_1 + 2\theta_2 - 12 = 0$$

Substitute $\theta_1 = \frac{-2\lambda}{5}$ and $\theta_2 = \frac{-\lambda}{3}$ into the constraint equation:

$$4\left(\frac{-2\lambda}{5}\right) + 2\left(\frac{-\lambda}{3}\right) = 12$$

Solving for λ :

$$\frac{-8\lambda}{5} + \frac{-2\lambda}{3} = 12 \quad \Rightarrow \quad \frac{-34\lambda}{15} = 12 \quad \Rightarrow \quad \lambda = \frac{-180}{34} = -\frac{90}{17}$$

Using the value of λ , we find:

$$\theta_1 = \frac{36}{17}, \quad \theta_2 = \frac{30}{17}$$

The optimal values of θ_1 and θ_2 are $\theta_1 = \frac{36}{17}$, $\theta_2 = \frac{30}{17}$, and the Lagrange multiplier is $\lambda = \frac{-90}{17}$.

Python Code Implementation

Below is the Python code to solve the optimization problem using the Lagrangian method and to visualize the contour plot with the constraint line:

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.optimize import minimize
4
5 # Define the cost function  $J(\theta) = 5\theta_1^2 + 3\theta_2^2$ 
6 def cost_function(theta):
7     return 5 * theta[0]**2 + 3 * theta[1]**2
8
9 # Define the constraint  $4\theta_1 + 2\theta_2 = 12$ 
10 def constraint_eq(theta):
11     return 4 * theta[0] + 2 * theta[1] - 12
12
13 # Initial guess for theta ( $\theta_1, \theta_2$ )
14 theta_initial = [0, 0]
15
16 # Define the constraint as a dictionary
17 constraint = {'type': 'eq', 'fun': constraint_eq}
18
19 # Use SciPy's minimize function to optimize with the constraint
20 solution = minimize(cost_function, theta_initial, constraints=[
21     constraint])
22
23 theta_opt = solution.x
24
25 # Extract the values of  $\theta_1$  and  $\theta_2$ 
26 theta_1_opt, theta_2_opt = theta_opt
27
28 # Print the results
29 print(f"Optimal  $\theta_1$ : {theta_1_opt}")
30 print(f"Optimal  $\theta_2$ : {theta_2_opt}")
31
32 # Prepare visualization: Contour plot and constraint line
33
34 # Generate a grid of values for  $\theta_1$  and  $\theta_2$ 
35 theta_1_values = np.linspace(-5, 5, 400)
36 theta_2_values = np.linspace(-5, 5, 400)
37 theta_1_grid, theta_2_grid = np.meshgrid(theta_1_values,
38     theta_2_values)
39
40 # Compute the cost function values for the grid
41 cost_grid = 5 * theta_1_grid**2 + 3 * theta_2_grid**2
42
43 # Plot the contour of the cost function
44 plt.contour(theta_1_grid, theta_2_grid, cost_grid, levels=20, cmap=
45     'viridis')
46
47 # Plot the constraint line  $4\theta_1 + 2\theta_2 = 12$ 
48 theta_2_constraint = (12 - 4 * theta_1_values) / 2
49 plt.plot(theta_1_values, theta_2_constraint, label="Constraint: 4
50      $\theta_1 + 2\theta_2 = 12$ ", color='r')
```

```

49 plt.scatter([theta_1_opt], [theta_2_opt], color='b', zorder=5,
    label="Optimal Solution")
50
51 # Labels and legend
52 plt.xlabel("theta1")
53 plt.ylabel("theta2")
54 plt.title("Contour Plot of Cost Function with Constraint Line")
55 plt.legend()
56
57 # Show the plot
58 plt.show()

```

